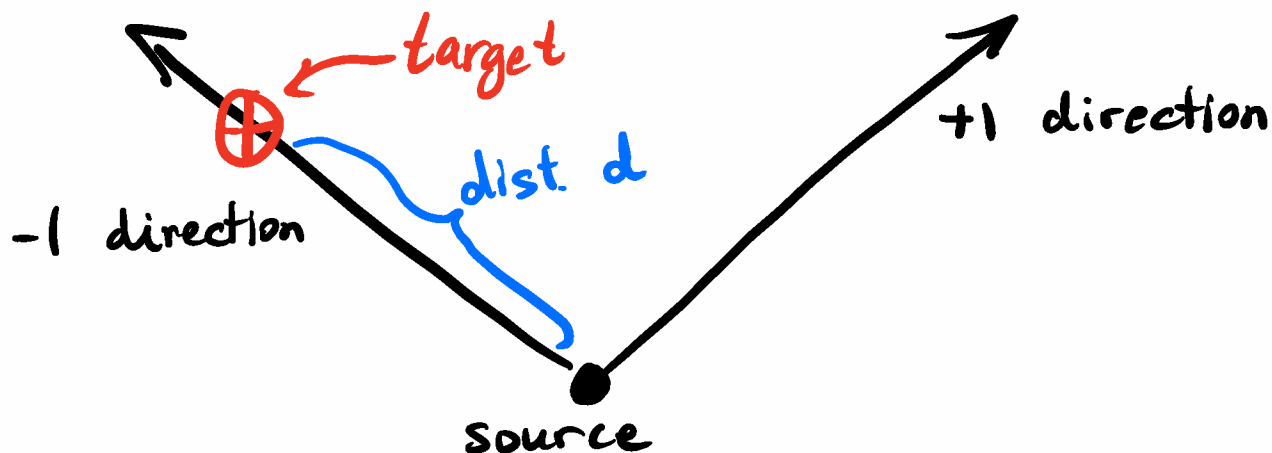


Simple examples:

1. Two roads diverge (in a yellow wood:)



We don't know if the target is to the left or to the right, and also we don't know d .

idea: guess d . use a sequence of guesses $1, 2, 4, 8, \dots, 2^i, \dots$

for guess 2^i move a distance of 2^i in direction $(-1)^i$ and if target not reached, go back to source.

what's the distance we traverse? -
the worst case is $d = 2^i + \epsilon$ and the target is in direction $(-1)^i$

in this case we pay:

$$\begin{aligned} 2 \cdot 1 + 2 \cdot 2 + 2 \cdot 4 + \dots + 2 \cdot 2^i + 2 \cdot 2^{i+1} + d &= \\ = 2 \cdot (2^{i+2} - 1) + 2^i + \epsilon &= 9 \cdot 2^i + \epsilon - 2 \approx 9 \cdot d \end{aligned}$$

knowing the location of the target, one pays d

2. paging:

computer memory is divided into n pages
fast access memory can hold $k < n$ pages

there is a sequence of page requests:

$$\sigma_1, \sigma_2, \sigma_3, \dots, \sigma_i, \dots \quad \sigma_i \in \{1, \dots, n\}$$

if σ_i is in fast memory (cache hit),
nothing needs to be done.

o.w. (cache miss), we need to bring σ_i
into fast memory, after evicting a page
that's already there to make room.
we pay 1 for every cache miss.

LRU - least recently used: evict the
page that was requested least recently.

partition the sequence into blocks:

$$B_1, B_2, \dots$$

B_i = longest prefix of the subsequence
excluding $B_1, \dots, B_{i-1}$ that contains
exactly k distinct pages

every block (except B_1) begins with
a request not seen in the previous block,

so any algorithm pays at least $\#blocks - 1$
LRU pay at most k times in a block
(whenever a new page is encountered)
overall cost $\leq k \cdot \#blocks$

claim: for every online algorithm there exists a sequence on which it pays $\geq k \cdot \text{opt.}$

proof: restrict the sequence to $k+1$ pages.
always request the page not in the algorithm's fast memory

so alg has $\#requests$ misses.

now consider k different algorithms.

let $\{1, 2, \dots, k\}$ be the initial config.

alg i starts by replacing i with $k+1$, the first request. from this point on, the "holes" of all k algs will be different and will include jointly just the last req.

upon receiving req. σ_j , only one of the k algs encounters a cache miss.

notice that σ_j is requested because alg evicted σ_j in favor of some other page σ_j' . so if alg i doesn't cover σ_j , it evicts σ_j' in favor of σ_j .

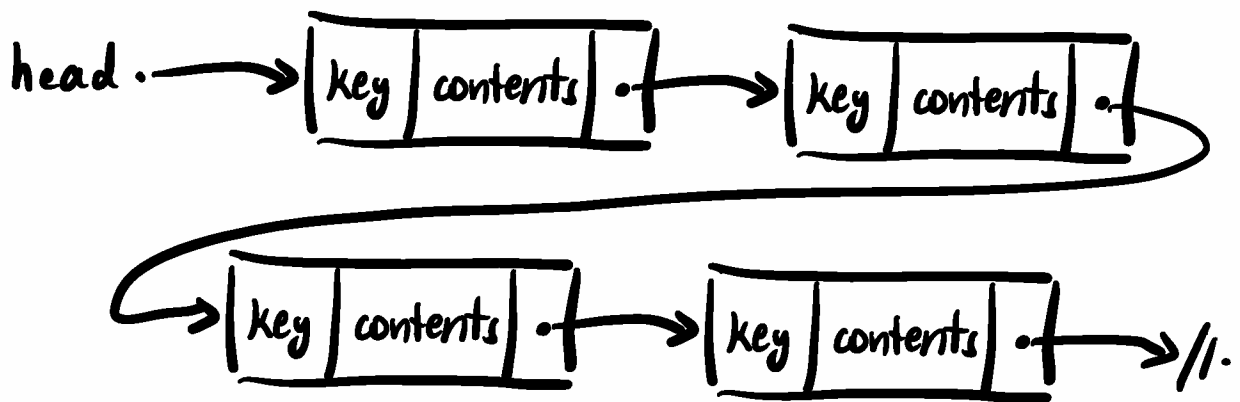
in the prev.
step

this maintains the invariant, as after the prer. step all k algs covered σ_j .

so the cost for all k algs together is $\# \text{requests} + k-1$, hence one of them for first req. pays $\leq \frac{\# \text{req.}}{k} + \frac{k-1}{k}$

3. Linked list update:

we have a linked list of n keyed entries (fixed set of items).



we get a sequence of search by key requests.

In response to a request, we must traverse the list, starting at the head, until we reach the requested target.

each step down the list costs 1

Then, we are also allowed to swap any two adjacent entries. There are two cost models:
option a: each swap costs 1
option b: swaps involving the searched entry cost 0, the rest cost 1.

we'll use option b here.

Move to Front (MTF): whenever an entry is searched, swap it backwards all the way to the front. (this costs 0 under option b)

a_i alg's current order

($a_i(j)$ = position of key j)

o_i optimal current order

C_i^{alg} = alg's cost in step i

$C_{i,\text{find}}^{\text{opt}}$ = opt cost in step i to find σ_i

$C_{i,\text{swap}}^{\text{opt}}$ = non-zero swap cost of opt in step i

$$C_i^{\text{opt}} = C_{i,\text{find}}^{\text{opt}} + C_{i,\text{swap}}^{\text{opt}}$$

$\phi(a, o) = \# \text{ inversions between } a \text{ \& } o$

(inversion: $a_i(j) < a_i(k)$ but $o_i(j) > o_i(k)$ or vice versa)

let's analyze $\phi(a_{i+1}, o_{i+1}) - \phi(a_i, o_i) \leq$

$$\leq \underbrace{\phi(a_{i+1}, o_i) - \phi(a_i, o_i)}_{\text{max positive change}} + \underbrace{\phi(a_{i+1}, o_{i+1}) - \phi(a_{i+1}, o_i)}_{\text{min negative change}}$$

$$\leq \underbrace{o_{i+1}(\sigma_{i+1}) - 1}_{\text{max positive change}} - \underbrace{(a_{i+1}(\sigma_{i+1}) - o_{i+1}(\sigma_{i+1}))}_{\text{min negative change}}$$

$$\leq C_{i,\text{swap}}^{\text{opt}}$$

$$\leq 2c_{r_{i+1}}(\sigma_{i+1}) + C_{i, \text{swap}}^{\text{opt}} - a_{i+1}(\sigma_{i+1})$$

$$\leq 2C_i^{\text{opt}} - C_i^{\text{alg}}$$

$$\sum_{i=1}^n C_i^{\text{alg}} \leq 2 \sum_{i=1}^n C_i^{\text{opt}} + \underbrace{\phi(a_0, o_0)}_{=0} - \underbrace{\phi(a_n, o_n)}_{\geq 0}$$

lower bound: for every deterministic online algorithm and for every $\epsilon > 0$ there is a sequence on which the algorithm pays at least $2-\epsilon$ time opt.

proof: the bad sequence requests always the last item in the current list. For T steps, this costs Tn .

We compare this performance against a static algorithm (not necessarily opt) which sorts the list in non-increasing frequency of request $P_1 \geq P_2 \geq \dots \geq P_n$ (i.e. we renumber the keys $\{1, \dots, n\}$ so that key i is requested $T \cdot P_i$ times, then sort the list in this order)

$$\text{cost} \leq n^2 + T \sum_{i=1}^n i P_i \leq n^2 + T \frac{n+1}{2}$$

an upper bound
on the cost of
sorting

maximized at $P_i = \frac{1}{n} \forall i$

The ratio approaches $2(1 - \frac{1}{n+1})$ as $T \rightarrow \infty$

questions:

1. What's the best you can do in a 3-way intersection of roads?
2. How good is FIFO paging? LIFO paging?
3. What's the performance of MTF if we count also swaps involving the searched item?

game theoretic model

two-player zero-sum game (a.k.a. request-answer games)

the players are:

the adversary (ADV)

the online player (ONL)

in strategic form:

Σ_{ADV} = set of actions (requests) of ADV

Σ_{ONL} = set of actions (answers) of ONL

ADV's pure strategies: $S_{ADV} = \{\sigma \mid \sigma \in \Sigma_{ADV}^*\}$
(request sequences)

ONL's pure strategies: $S_{ONL} = \{\alpha \mid \alpha: \Sigma_{ADV}^* \rightarrow \Sigma_{ONL}\}$
(answer sequences $\alpha(\sigma_1) \alpha(\sigma_1, \sigma_2) \alpha(\sigma_1, \sigma_2, \sigma_3) \dots$)

we associate a cost (or profit, in the case of maximization problems) with each play (i.e., pair σ, α): $C: S_{ADV} \times S_{ONL} \rightarrow \mathbb{R}_+ \cup \{\infty\}$

we also have a base cost:

$$C_{base}: S_{ADV} \rightarrow \mathbb{R}_+ \cup \{\infty\}$$

$$\text{for now: } C_{base}(\sigma) = C_{opt}(\sigma) = \inf_{\alpha \in S_{ONL}} C(\sigma, \alpha)$$

example: for the paging problem with n pages and cache capacity k -

$$\Sigma_{ADV} = \{1, 2, \dots, n\} \quad \Sigma_{ONL} = \{k \text{ distinct elements } \in \Sigma_{ADV}\}$$

$$C(\sigma, \alpha) = \sum_{t=1}^{|\sigma|} \text{cost}(\sigma_t, \alpha(\sigma_1 \dots \sigma_{t-1}), \alpha(\sigma_1, \dots, \sigma_t)), \text{ where}$$

$$\text{cost}(s, A_0, A_1) = \begin{cases} \infty & s \notin A_1 \\ k - |A_0 \cap A_1| & \text{o.w.} \end{cases}$$

$\alpha(\epsilon) = \text{initial config.}$

↖ empty sequence

↖ cost of switching from A_0 to A_1

def: Let α be a deterministic online algorithm (pure ONL strategy). The competitive ratio of α is $C_\alpha = \inf \{C \mid \exists A \forall \sigma \in \Sigma_{ADV} C(\sigma, \alpha) \leq C \cdot C_{\text{case}}(\sigma) + A\}$

(def for minimization problem)

The competitive ratio of the game is

$$C_{\text{game}}^{\text{det}} = \inf_{\alpha \in \Sigma_{ONL}} C_\alpha$$

example: we proved that $C_{\text{paging}(n,k)}^{\text{det}} = k$

LRU gives the upper bound.

we also proved that $C_{\text{list-upd}(n)}^{\text{det}} = 2$

MTF gives the upper bound.

metrical task systems (MTS)

There is a finite state space X
and a metric $d: X \times X \rightarrow \mathbb{R}$:

$$\forall x, y \in X, d(x, y) \geq 0$$

$$d(x, y) = 0 \iff x = y$$

$$\forall x, y \in X, d(x, y) = d(y, x)$$

$$\forall x, y, z \in X, d(x, y) + d(y, z) \geq d(x, z)$$

$$\Sigma_{\text{ADV}} = \{ \sigma: X \rightarrow \mathbb{R}_+ \cup \{\infty\} \}$$

$\sigma \in \Sigma$ assigns to each $x \in X$ a service cost $\sigma(x)$.

$$\Sigma_{\text{ONL}} = X$$

$$c(\sigma, \alpha) = \sum_{t=1}^{|\sigma|} \left[d(\alpha(\sigma_1 \dots \sigma_{t-1}), \alpha(\sigma_1 \dots \sigma_t)) + \sigma(\alpha(\sigma_1 \dots \sigma_t)) \right]$$

movement cost

service cost

we'll scale d such that $\min_{i \neq j} d(i, j) = 1$.

$$\text{diam}(X, d) = \max_{i, j} d(i, j)$$

models many problems -

paging: $X = \binom{n}{k}$ possible contents of cache

$$d(x, y) = k - |x \cap y|$$

$$\sigma_t(x) = \begin{cases} 0 & \text{if requested page in step } t \in x \\ \infty & \text{o.w.} \end{cases}$$

(if service costs $\in \{0, \epsilon\}$ this is called a metrical service system)

linked list update (with option (a) costs):

X = all permutations on $\{1, \dots, n\}$

$d(x, y)$ = #swaps to convert x into y

$\sigma_t(x)$ = location of t -th request in x .

elementary cost vectors

$$e_i = (0, 0, \dots, 0, \epsilon, 0, 0, \dots, 0)$$

↑
 i -th position

$$i \in X \quad \epsilon > 0 \text{ tiny}$$

fact: it is sufficient to consider requests which are elementary cost vectors

Thm: $C_{\text{MTS}(n)}^{\det} \geq 2n - 1$

proof: fix a deterministic online algorithm α .

the adversary requests at step t $\sigma_t = e_{i_{t-1}}$, where

$i_t = \alpha(\sigma_1, \sigma_2, \dots, \sigma_t)$ is the state of α after step t .

consider a time horizon T and suppose that in T steps α changes state k times.

$$\text{then, } C_\alpha(\sigma_1, \dots, \sigma_T) = (T - k)\epsilon + \sum_{t=1}^T d(i_{t-1}, i_t) \geq k\epsilon$$

we define $2n - 1$ other algorithms:

$$\alpha_1, \alpha_2, \dots, \alpha_{2n-1}$$

at time t , there are two algorithms that occupy each state $i \in X \setminus \{i_t\}$ and one algorithm that occupies state i_t .
 the cost of initiating this invariant at time $t=0$ is $\leq 2(n-1) \cdot \text{diam}(X, d)$

if $i_{t-1} = i_t$, it costs nothing to maintain the invariant. o.w., the algorithm pays $d(i_{t-1}, i_t)$, and we move one of the two algorithms at i_t to i_{t-1} , so we pay $d(i_t, i_{t-1}) = d(i_{t-1}, i_t)$.

Note also that exactly one algorithm pays ε to serve σ_t , and the rest pay 0.

$$\text{Thus: } \sum_{r=1}^{2n-1} C_{\alpha_r}(\sigma_1, \sigma_2, \dots, \sigma_T) \leq \\ \leq T\varepsilon + \sum_{t=1}^T d(i_{t-1}, i_t) + 2(n-1) \cdot \text{diam}(X, d)$$

so the best algorithm pays at most $\frac{1}{2n-1} C_{\alpha}(\sigma_1, \dots, \sigma_T) + \frac{2n-2}{2n-1} \cdot \text{diam}(X, d) + \frac{k\varepsilon}{2n-1}$

if $\varepsilon \rightarrow 0$ and $T\varepsilon \rightarrow \infty$, the first term dominates the sum.

work function (WF)

$$W_t: X \rightarrow \mathbb{R}_+$$

$W_t(x)$ = minimum cost of serving $\sigma_1 \sigma_2 \dots \sigma_t$ and ending in state x .

dynamic programming:

$$W_0(x) = \begin{cases} 0 & \text{if } x \text{ is the initial state } x_0 \\ d(x_0, x) & \text{o.w.} \end{cases}$$

$$W_{t+1}(x) = \min_{y \in X} \{W_t(y) + \sigma_{t+1}(y) + d(y, x)\} \quad \begin{matrix} \forall t \geq 0 \\ \forall x \in X \end{matrix}$$

given d, W_t, σ_{t+1} computing W_{t+1} takes $O(|X|^2)$ arithmetic operations

$$\text{Copt}(\sigma_1 \sigma_2 \dots \sigma_T) = \min_{x \in X} W_T(x)$$

claim: $\forall x, y \in X \quad \forall t, W_t(y) \leq W_t(x) + d(x, y)$

$$\begin{aligned} \text{proof: } W_t(y) &= \min_{z \in X} \{W_{t-1}(z) + \sigma_t(z) + d(z, y)\} \leq \\ &\leq \min_{z \in X} \{W_{t-1}(z) + \sigma_t(z) + d(x, z)\} + d(y, x) \end{aligned}$$

triangle inequality $\Rightarrow W_t(x) + d(x, y)$

$$d(y, z) \leq d(y, x) + d(x, z)$$

$$\forall z \Rightarrow \forall x \in X, W_T(x) \leq \text{Copt}(\sigma_1 \dots \sigma_T) + \text{diam}(X, d)$$

if the claim holds with equality, we say that y is dominated by x at time t . Note that if $\sigma_{t+1} = e_y$, then $w_{t+1}(y) > w_t(y)$ if y is not dominated by any x at time t .

work function algorithm (WFA)

$\forall t \geq 1$, serve σ_t at $x_t = \arg \min_{x \in X} \{w_t(x) + d(x_{t-1}, x)\}$

Thm: WFA is $(2n-1)$ -competitive

proof: define a potential function

$$\phi(t) = 2 \sum_{x \in X \setminus \{x_t\}} w_t(x) + w_t(x_t)$$

the cost of WFA at step t is

$$C_{WFA}(t) = d(x_{t-1}, x_t) + \sigma_t(x_t)$$

we will argue that $C_{WFA}(t) \leq \phi(t) - \phi(t-1)$

case 1: $C_{WFA}(t) = 0$. trivial, as $w_t(x)$ is monotonically non-decreasing in t , for all x .

if $C_{WFA}(t) > 0$ it must be that $\sigma_t = e_{x_{t-1}}$

case 2: $x_{t-1} = x_t$. it must be that x_{t-1} is not dominated by any x , so $w_t(x_{t-1}) = w_{t-1}(x_{t-1}) + \epsilon$

and therefore $\phi(t) = \phi(t-1) + \epsilon$

case 3: $x_{t-1} \neq x_t$. in this case x_{t-1} must be dominated by x_t at time $t-1$ (i.e., as ϵ is tiny, it's better not to move). So, $w_t = w_{t-1}$.

$$\phi(t) - \phi(t-1) = w_t(x_{t-1}) - w_t(x_t) = d(x_{t-1}, x_t)$$

Thus,

$$C_{WFA}(\sigma_1 \sigma_2 \dots \sigma_T) = \sum_{t=1}^T C_{WFA}(t) = \phi(T) - \phi(0) \leq \\ \leq \phi(T) \leq (2n-1) C_{OPT}(\sigma_1 \sigma_2 \dots \sigma_T) + \underbrace{(2n-2) \text{diam}(X, d)}_{\text{const.}}$$

a more general model

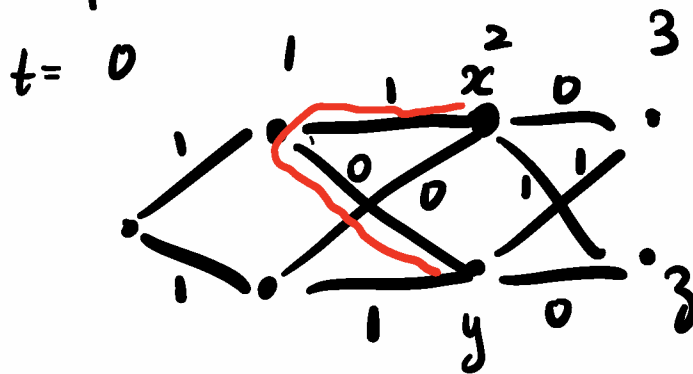
at every step t , a new state space X_t is revealed, along with a distance function

$d_t: X_{t-1} \times X_t \rightarrow \mathbb{R}_+$ extending the metric

so far. I.e., $\forall x, y \in X_{t-1}, \forall z \in X_t$,

$d_t(x, z) \leq d(x, y) + d_t(y, z)$, where $d(x, y)$ is

the shortest path between x and y through $\bigcup_{j=0}^{t-1} X_j$



$$1 = d(x, z) \leq d(x, y) + d_t(y, z) = 1 + 0$$

layered graph traversal (a.k.a. small set chasing)

questions:

1. Argue why linked list update with option (b) costs (swaps using searched item cost 0) is not a metrical task system. Model it as a request-answer game.
2. Prove that if an MTS algorithm α is C -competitive againsts sequences of elementary cost vectors, then it is C -competitive against any sequence.
3. Show that layered graph traversal where $\forall t$ $|X_t| \leq 2$ has a deterministic 9-competitive online algorithm. You may assume that the minimum non-zero distance is 1.

Setting: a metric space $\mathcal{M} = (X, d)$
set of points $\rightarrow$ X metric on X $\rightarrow$ d

k servers, each occupying a point in X
the servers are indistinguishable other than location

ADV strategies: sequences $\sigma_1, \sigma_2, \sigma_3, \dots$ of points in X

ONL strategies: sequences $x^{(1)} x^{(2)} \dots$ of configurations
s.t. $\sigma_t \in X^{(k)}$ in X^k ↗

$x^{(0)}$ = initial configuration

$$\text{cost: } \sum_t \min_{\pi} \sum_{i=1}^k d(x_i^{(t-1)}, x_{\pi(i)}^{(t)})$$

↗
permutation

there's no need to place more than one server at a point, and no need to move more than one server, to σ_t , per request (but these constraints may require extra memory)

example: paging - $|X| = n$

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & \text{o.w.} \end{cases}$$

weighted caching - $p: X \rightarrow \mathbb{R}_+$

$$d(x, y) = \frac{1}{2}(p(x) + p(y))$$

If $|X| = n$ then k -server is a metrical service system on $\binom{n}{k}$ states. But we seek a competitive ratio that doesn't depend on n .

If $|X| = k+1$ there are $k+1$ states $\Rightarrow$ c.r. $\leq 2k+1$

WFA for metrical task ↗ systems

lower bound: $\forall \mathcal{M} = (X, d)$ with $|X| \geq k+1$

$\forall$ ONL strategy α , $C_\alpha \geq k$.

proof: Let $\{1, 2, \dots, k+1\} \subseteq X$ include the points in $x^{(0)}$. Consider the ADV strategy: $\forall t$, σ_t is the unique point in $\{1, \dots, k+1\} \setminus \{x^{(t-1)}\}$. After T steps:
$$C_\alpha(\sigma) \geq \sum_{t=1}^{T-1} d(\sigma_{t+1}, \sigma_t) \quad (\text{ignoring the last step})$$

Consider k distinct algorithms that maintain the following invariant $\forall t \geq 1$:

- (i) they all occupy the last request σ_t
- (ii) the "hole" (point with no server) of each alg. is distinct

For $t=1$, each alg. moves a different server to σ_1

For $t > 1$, the unique alg. with hole at σ_t (by induction) moves a server from σ_{t-1} to σ_t

$$\begin{aligned} \text{total cost} &= \sum_{i \neq \sigma_1} d(i, \sigma_1) + \sum_{t=2}^T d(\sigma_{t-1}, \sigma_t) \\ &\leq C_\alpha(\sigma) + k \cdot \text{diam}(\{1, \dots, k+1\}, d) \end{aligned}$$

So, $\exists$ alg. β s.t. $\forall T$

$$C_\beta(\sigma_1 \dots \sigma_T) \leq \frac{1}{k} C_\alpha(\sigma_1 \dots \sigma_T) + \text{diam}(\dots)$$

$$\Rightarrow \underbrace{C_\alpha(\sigma_1 \dots \sigma_T)}_{\substack{\rightarrow \infty \\ T \rightarrow \infty}} \geq k \cdot C_\beta(\sigma_1 \dots \sigma_T) - \underbrace{k \cdot \text{diam}(\dots)}_{\text{const.}}$$

Thm: If $|X| = k+1$ then $C_{k\text{-srv}}(X, d) = k$.

proof: later ...

conjecture: If $|X| \geq k+1$ then $C_{k\text{-srv}}(X, d) = k$.

k -server on the real line: $x_1 \leq x_2 \leq \dots \leq x_k$ alg config.
 $y_1 \leq y_2 \leq \dots \leq y_k$ opt config

double coverage algorithm (DCA):

$$\sigma_t \in (-\infty, x_1^{(t-1)})$$

move server at $x_1^{(t-1)}$ to σ_t

$$\sigma_t \in (x_k^{(t-1)}, \infty)$$

move server at $x_k^{(t-1)}$ to σ_t

$$\sigma_t \in (x_i^{(t-1)}, x_{i+1}^{(t-1)})$$

move both servers at $x_i^{(t-1)}, x_{i+1}^{(t-1)}$ towards σ_t a distance of

$$\min\{|x_i^{(t-1)} - \sigma_t|, |x_{i+1}^{(t-1)} - \sigma_t|\}$$

o.w.

no move needed

Thm: $C_{DCA} = k$

proof: Consider the following potential function:

$$\phi(x, y) = k \sum_{i=1}^k |x_i - y_i| + \sum_{1 \leq i < j \leq k} |x_i - x_j|$$

at time t , ADV requests σ_t .

OPT moves from $y_{i_t}^{(t-1)}$ to σ_t paying $|y_{i_t}^{(t-1)} - \sigma_t|$

$$\phi(x^{(t-1)}, y^{(t)}) - \phi(x^{(t-1)}, y^{(t-1)}) =$$

$$= k(|x_{i_t}^{(t-1)} - \sigma_t| - |x_{i_t}^{(t-1)} - y_{i_t}^{(t-1)}|) \leq$$

$$\leq k |y_{i_t}^{(t-1)} - \sigma_t|$$

$$\phi(x^{(t)}, y^{(t)}) - \phi(x^{(t-1)}, y^{(t)}) =$$

$$\leq \begin{cases} -k|\sigma_t - x_1^{(t-1)}| + \underbrace{\sum_{i=2}^k |\sigma_t - x_i^{(t-1)}|}_{\text{dist. from } x_1 \text{ to rest increases}} & \sigma_t < x_1^{(t-1)} \\ & \parallel x_1^{(t)} \\ & \text{as } y_1^{(t)} \leq \sigma_t \\ \\ -k|\sigma_t - x_k^{(t-1)}| + \sum_{i=1}^{k-1} |\sigma_t - x_i^{(t-1)}| & \sigma_t > x_k^{(t-1)} \\ & \text{(Similar analysis)} \end{cases}$$

$$0 - 2 \cdot \min\{|\sigma_t - x_i^{(t-1)}|, |\sigma_t - x_{i+1}^{(t-1)}|\}$$

there an OPT server at σ_t so it can't be that both servers at $x_i^{(t-1)}, x_{i+1}^{(t-1)}$ move away from $y_i^{(t)}, y_{i+1}^{(t)}$ respectively

the two servers move towards each other and cancel each other's effect on dists to other servers

$\leq$ - cost of DCA in step t

$$\Rightarrow -\sum_{1 \leq i < j \leq k} |x_i^{(0)} - x_j^{(0)}| \leq \phi(x^{(T)}, y^{(T)}) - \phi(x^{(0)}, y^{(0)}) \leq \\ \leq k C_{OPT}(\sigma_1 \dots \sigma_T) - C_{DCA}(\sigma_1 \dots \sigma_k)$$

work function: $\mathcal{M}=(X, d)$ for simplicity assume X is finite $|X|=n$.

sequence of requests $\sigma_1, \sigma_2, \sigma_3 \dots \in X$
initial configuration $x_1^{(0)}, x_2^{(0)}, \dots, x_k^{(0)}$

$$W_t: X^k \rightarrow [0, \infty)$$

$W_t(x_1, \dots, x_k)$ = minimum cost to serve $\sigma_1, \sigma_2, \dots, \sigma_t$ starting at $x^{(0)}$ and ending at x .

computable in time $t n^{O(k)}$ using dynamic prog.

$$W_t(x) = \begin{cases} \min_{x'} W_{t-1}(x') + d(x, x') & \sigma_t \in x \\ \min_{\substack{x' \\ x' \ni \sigma_t}} W_t(x') + d(x', x) & \sigma_t \notin x \end{cases}$$

where $d(x, x') = \min_{\text{perm. } \pi} \sum_{i=1}^k d(x_i, x'_{\pi(i)})$

base case:

$$W_0(x^{(0)}) = 0, \forall x \neq x^{(0)}, W_0(x) = d(x, x^{(0)})$$

work function algorithm WFA

in response to σ_t move to $x^{(t)} (\ni \sigma_t)$

that minimizes $W_t(x) + d(x, x^{(t-1)})$

questions:

1. Prove Fact 1:

We can choose for WFA configurations $x^{(1)}, x^{(2)}, \dots, x^{(t)}$ for which $\forall t=1, 2, \dots$

$$|\{x_1^{(t)}, \dots, x_k^{(t)}\} \setminus \{x_1^{(t-1)}, \dots, x_k^{(t-1)}\}| \leq 1$$

2. Prove that $\forall t=1, 2, \dots$ and for all $x \in X^k$,

$$W_t(x) = \min_{i=1, \dots, k} \left\{ \underbrace{W_{t-1}(x_1, \dots, x_{i-1}, \sigma_t, x_{i+1}, \dots, x_k)}_{= W_t(x_1, \dots, x_{i-1}, \sigma_t, x_{i+1}, \dots, x_k)} + d(\sigma_t, x_i) \right\}$$

The work function algorithm for k-server

work function: $\mathcal{M}=(X,d)$ for simplicity assume X is finite $|X|=n$.

sequence of requests $\sigma_1, \sigma_2, \sigma_3 \dots \in X$
initial configuration $x_1^{(0)}, x_2^{(0)}, \dots, x_k^{(0)}$

$$W_t: X^k \rightarrow [0, \infty)$$

$W_t(x_1, \dots, x_k)$ = minimum cost to serve $\sigma_1, \sigma_2, \dots, \sigma_t$ starting at $x^{(0)}$ and ending at x .

computable in time $t n^{O(k)}$ using dynamic prog.

$$W_t(x) = \begin{cases} \min_{x'} W_{t-1}(x') + d(x, x') & \sigma_t \in x \\ \min_{x' \ni \sigma_t} W_t(x') + d(x', x) & \sigma_t \notin x \end{cases}$$

where $d(x, x') = \min_{\text{perm. } \pi} \sum_{i=1}^k d(x_i, x'_{\pi(i)})$

base case:

$$W_0(x^{(0)}) = 0, \forall x \neq x^{(0)}, W_0(x) = d(x, x^{(0)})$$

work function algorithm WFA

in response to σ_t move to $x^{(t)} (\ni \sigma_t)$

that minimizes $W_t(x) + d(x, x^{(t-1)})$

Thm: If $n = k+1$ then WFA is k -competitive.

proof: We'll use (homework assignment):

fact 1: We can choose $x^{(t)}$ that differs from $x^{(t-1)}$ in the location of at most one server
 $(|\{x_1^{(t)}, \dots, x_k^{(t)}\} \setminus \{x_1^{(t-1)}, \dots, x_k^{(t-1)}\}| \leq 1)$

def: the extended cost at step t is

$$C_t = \max_{x \in X^k} (W_t(x) - W_{t-1}(x)).$$

fact 2: If $\exists a$ s.t. $\forall \sigma$

$\sum_{t=1}^m C_t \leq (c+1) C_{\text{base}}(\sigma) + a$, then WFA is c -competitive.

proof: (later)

def: $x \in X^k$ is a minimizer w.r.t. $r \in X$ and W_t iff, $W_t(x) - \sum_{i=1}^k d(r, x_i) = \min_{x' \in X^k} W_t(x') - \sum_{i=1}^k d(r, x'_i)$

Lemma (duality): Any minimizer x w.r.t. σ_t and W_{t-1} is also a minimizer w.r.t. σ_t and W_t , and the extended cost of step t is $W_t(x) - W_{t-1}(x)$.

proof (of thm): For $r \in X$ let $x^{(t,r)}$ denote a minimizer configuration w.r.t. r and W_t .

Define $\phi_t = \sum_{r \in X} (W_t(x^{(t,r)}) - \sum_{i=1}^k d(r, x_i^{(t,r)}))$

$\phi_0 = \text{const.}$

Lemma: (1) $\phi_{t+1} - \phi_t \geq C_{t+1}$

(2) $\phi_t - n \cdot \min_{y \in X^k} W_t(y) \leq \text{const.}$

OPT

proof: (1) Notice that $x^{(t, \sigma_{t+1})}$ is a minimizer config. w.r.t. σ_{t+1} and both W_t and W_{t+1} . Thus, $C_{t+1} = W_{t+1}(x^{(t, \sigma_{t+1})}) - W_t(x^{(t, \sigma_{t+1})})$

This accounts for the term for σ_{t+1} in $\phi_{t+1} - \phi_t$.

If $r \neq \sigma_{t+1}$ then

$$\begin{aligned} & (W_{t+1}(x^{(t+1,r)}) - \sum_{i=1}^k d(r, x_i^{(t+1,r)})) - (W_t(x^{(t,r)}) - \sum_{i=1}^k d(r, x_i^{(t,r)})) \geq \\ & \geq (W_{t+1}(x^{(t+1,r)}) - \sum_{i=1}^k d(r, x_i^{(t+1,r)})) - (W_t(x^{(t+1,r)}) - \sum_{i=1}^k d(r, x_i^{(t+1,r)})) = \\ & = W_{t+1}(x^{(t+1,r)}) - W_t(x^{(t+1,r)}) \geq 0. \end{aligned}$$

(2) Notice that $\forall x, W_t(x) \leq \min_y W_t(y) + k \cdot \text{diam}(\mathcal{U})$

Thus, $\phi_t \leq n \cdot \min_y W_t(y) + nk \cdot \text{diam}(\mathcal{U})$

back to the proof of the thm:

$$n \cdot \min_y W_m(y) \geq \phi_m - \text{const.} = \sum_{t=1}^m (\phi_t - \phi_{t-1}) + \phi_0 - \text{const.} \geq \sum_{t=1}^m C_t + (\phi_0 - \text{const.})$$

$\Rightarrow$ WFA is $(n-1)$ -competitive

fact 2

proof of fact 2:

(from homework assignment)

$$W_{t+1}(x) = \min_{i \in \{1, \dots, k\}} \left\{ W_t(x_1, \dots, x_{i-1}, \sigma_{t+1}, x_{i+1}, \dots, x_k) + d(\sigma_{t+1}, x_i) \right\}$$

$= W_{t+1}(\dots)$

Clearly, for any work function W , $\forall$ configs x, y

$$W(x) \leq W(y) + d(x, y)$$

$$\Rightarrow \forall t, \forall x, W_{t+1}(x) \geq W_t(x)$$

Now, suppose that $x_j^{(t+1)} = \begin{cases} x_j^{(t)} & j \neq i \\ \sigma_{t+1} & \text{o.w.} \end{cases}$

$$\begin{aligned} C_{t+1} &= \max_x \{ W_{t+1}(x) - W_t(x) \} \geq W_{t+1}(x^{(t+1)}) - W_t(x^{(t+1)}) \\ &= W_{t+1}(x^{(t+1)}) - W_t(x^{(t)}) + d(\sigma_{t+1}, x_i^{(t)}) \end{aligned}$$

So, if

$$(C+1) C_{\text{base}}(\sigma) + a \geq \sum_{t=1}^{|G|} C_t \geq \sum_{t=1}^{|G|} (W_t(x^{(t+1)}) - W_{t-1}(x^{(t+1)})) +$$

$$+ d(\sigma_t, x_{i_{t-1}}^{(t-1)}) = W_{b1}(x^{(b1)}) - W_0(x^{(0)}) +$$

$$C_{WFA}(\sigma) \geq C_{base}(\sigma) + C_{WFA}(\sigma)$$

$$\Rightarrow C_{WFA}(\sigma) \leq c \cdot C_{base}(\sigma) + a$$

proving fact 2.

quasi-convexity:

definition: $f: X^k \rightarrow [0, \infty)$ is quasi-convex

iff $\forall x, y \in X^k \exists h: \{1, \dots, k\} \leftrightarrow \{1, \dots, k\}$ s.t.

$\forall$ partitions S, T of $\{1, \dots, k\}$,

$f(x) + f(y) \geq f(x') + f(y')$, where

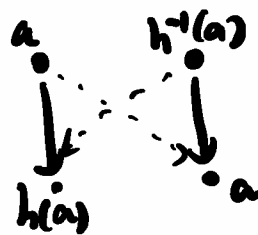
$$x'_j = \begin{cases} x_j & j \in S \\ y_{h(j)} & j \in T \end{cases} \quad y'_j = \begin{cases} y_{h(j)} & j \in S \\ x_j & j \in T \end{cases}$$

Lemma: It is sufficient to consider bijections h which are the identity on points that appear both in x and in y (let's regard x, y as multisets of their entries)

proof: Let h satisfying the def. maximize the #points in $x \cap y$ that are mapped to themselves. Now, assume that there is $a \in x \cap y$ s.t. $h(a) \neq a$

define

$$h'(b) = \begin{cases} a & b=a \\ h(a) & b=h^{-1}(a) \\ h(b) & \text{o.w.} \end{cases}$$



Consider a bipartition S, T of $\{x_1, \dots, x_k\}$

W.l.o.g., $h^{-1}(a) \in S$. If $a \in S$ then $h(S) = h'(S)$

and $h(T) = h'(T)$ so h' satisfies the def.

o.w. the condition for S, T can be derived

from the condition for $S \cup \{a\}, T \setminus \{a\}$ as

follows: $h(S \cup \{a\}) = h'(S \cup \{a\})$ and $h(T \setminus \{a\}) = h'(T \setminus \{a\})$

$$\text{So, } S \cup \{a\} \cup h(T \setminus \{a\}) = S \cup \{a\} \cup h'(T \setminus \{a\}) =$$

$$= S \cup \{a\} \cup h'(T) \text{ (as } h^{-1}(a) = a) \text{ and also}$$

$$h(S \cup \{a\}) \cup T \setminus \{a\} = h'(S \cup \{a\}) \cup T \setminus \{a\} =$$

$$= h'(S) \cup T$$

Lemma: $\forall t, W_t$ is quasi-convex.

proof: by induction on t .

$$\text{base case: } t=0. W_0(x) + W_0(y) = d(x^{(0)}, x) + d(x^{(0)}, y)$$

Let π_x, π_y be the permutations on $\{1, \dots, k\}$ that determine the two distances. ($\pi_x: x \rightarrow x^{(0)}, \pi_y: y \rightarrow x^{(0)}$)

Set $\forall a \in x, h(a) = \pi_y^{-1}(\pi_x(a))$. For any bipartition

$$S, T \text{ of } x, W_0(S \cup h(T)) + W_0(h(S) \cup T) = d(x^{(0)}, S \cup h(T)) + d(x^{(0)}, h(S) \cup T) = (\text{for two minimizing permutations})$$

$$\begin{aligned}
& \sum_{s \in S} d(s, \alpha(s)) + \sum_{s' \in h(T)} d(s', \alpha(s')) + \alpha, \beta: \rightarrow x^{(\omega)} \\
& + \sum_{s'' \in h(S)} d(s'', \beta(s'')) + \sum_{s''' \in T} d(s''', \beta(s''')) \leq \\
& \leq \sum_{s \in S \cup T} d(s, \pi_x^{-1}(s)) + \sum_{s' \in h(S) \cup h(T)} d(s', \pi_y^{-1}(s')) = \\
& = W_0(x) + W_0(y).
\end{aligned}$$

Inductive step: assume W_t is quasi-convex.

for some $a \in x$, $W_{t+1}(x) = W_t(x \setminus \{a\} \cup \{\sigma_{t+1}\}) + d(\sigma_{t+1}, a)$

for some $b \in y$, $W_{t+1}(y) = W_t(y \setminus \{b\} \cup \{\sigma_{t+1}\}) + d(\sigma_{t+1}, b)$

By the induction hypothesis, $\exists h: x \setminus \{a\} \cup \{\sigma_{t+1}\} \leftrightarrow y \setminus \{b\} \cup \{\sigma_{t+1}\}$ that satisfies the def. Moreover, $h(\sigma_{t+1}) = \sigma_{t+1}$.

define $h'(l) = \begin{cases} b & l = a \\ h(l) & \text{o.w.} \end{cases}$

let S, T be a bipartition of x , w.l.o.g. $a \in S$.

$$\begin{aligned}
W_{t+1}(x) + W_{t+1}(y) &= W_t(x \setminus \{a\} \cup \{\sigma_{t+1}\}) + W_t(y \setminus \{b\} \cup \{\sigma_{t+1}\}) \\
&+ d(\sigma_{t+1}, a) + d(\sigma_{t+1}, b) =
\end{aligned}$$

$$\begin{aligned}
&= W_t(S \setminus \{a\} \cup \{\sigma_{t+1}\} \cup T) + W_t(y \setminus \{b\} \cup \{\sigma_{t+1}\}) + d(\sigma_{t+1}, a) + \\
&+ d(\sigma_{t+1}, b) \geq
\end{aligned}$$

$$\begin{aligned}
&W_t(S \setminus \{a\} \cup \{\sigma_{t+1}\} \cup h(T)) + W_t(h(S \setminus \{a\} \cup \{\sigma_{t+1}\}) \cup T) \\
&+ d(\sigma_{t+1}, a) + d(\sigma_{t+1}, b) =
\end{aligned}$$

$$\begin{aligned}
&= W_t(S \setminus \{a\} \cup \{\sigma_{t+1}\} \cup h'(T)) + W_t(h'(S \setminus \{a\} \cup \{\sigma_{t+1}\}) \cup T) \\
&\quad + d(\sigma_{t+1}, a) + d(\sigma_{t+1}, b) \geq \\
&W_{t+1}(S \cup h'(T)) + W_{t+1}(h'(S) \cup T)
\end{aligned}$$

conclusion: $\forall a \in x \exists b \in y, W_t(x) + W_t(y) \geq W_t(x \setminus \{a\} \cup \{b\}) + W_t(y \setminus \{b\} \cup \{a\})$

proof of duality lemma, part 1:

let x be a minimizer of σ_t w.r.t. W_{t-1} .

It suffices to prove that $\forall y$

$$W_t(y) - \sum_{b \in y} d(\sigma_t, b) \geq W_t(x) - \sum_{a \in x} d(\sigma_t, a) \quad \text{or}$$

$$W_t(y) - \sum_{b \in y} d(\sigma_t, b) + W_{t-1}(x) \geq W_t(x) - \sum_{a \in x} d(\sigma_t, a) + W_{t-1}(x)$$

$$\exists b' \in y \text{ s.t. } W_t(y) = W_{t-1}(y \setminus \{b'\} \cup \{\sigma_t\}) + d(\sigma_t, b')$$

by quasiconvexity, $\exists a' \in x$ s.t.

$$\begin{aligned}
W_{t-1}(y \setminus \{b'\} \cup \{\sigma_t\}) + W_{t-1}(x) &\geq W_{t-1}(y \setminus \{b'\} \cup \{a'\}) + \\
&\quad + W_{t-1}(x \setminus \{a'\} \cup \{\sigma_t\})
\end{aligned}$$

by the assumption on x ,

$$w_{t-1}(y \setminus \{b'\} \cup \{a'\}) - \sum_{b \in y \setminus \{b'\} \cup \{a'\}} d(\sigma_t, b) \geq$$

$$w_{t-1}(x) - \sum_{a \in x} d(\sigma_t, a)$$

$$\text{so, } w_t(y) + w_{t-1}(x) - \sum_{b \in y} d(\sigma_t, b) =$$

$$= w_{t-1}(y \setminus \{b'\} \cup \{\sigma_t\}) + d(\sigma_t, b') + w_{t-1}(x) - \sum_{b \in y} d(\sigma_t, b)$$

$$= w_{t-1}(y \setminus \{b'\} \cup \{\sigma_t\}) + w_{t-1}(x) - \sum_{b \in y \setminus \{b'\} \cup \{\sigma_t\}} d(\sigma_t, b) \geq$$

$$\geq w_{t-1}(y \setminus \{b'\} \cup \{a'\}) + w_{t-1}(x \setminus \{a'\} \cup \{\sigma_t\}) -$$

$$- \sum_{b \in y \setminus \{b'\} \cup \{\sigma_t\}} d(\sigma_t, b) =$$

$$= w_{t-1}(y \setminus \{b'\} \cup \{a'\}) + w_{t-1}(x \setminus \{a'\} \cup \{\sigma_t\}) +$$

$$+ d(\sigma_t, a') - \sum_{b \in y \setminus \{b'\} \cup \{a'\}} d(\sigma_t, b) \geq$$

$$\geq w_{t-1}(x) - \sum_{a \in x} d(\sigma_t, a) + w_{t-1}(x \setminus \{a'\} \cup \{\sigma_t\}) +$$

$$+ d(\sigma_t, a') \geq w_{t-1}(x) - \sum_{a \in x} d(\sigma_t, a) + w_t(x)$$

proof of duality lemma, part 2:

x is a minimizer of σ_t w.r.t. W_{t-1} (and W_t).

It suffices to show, for all y ,

$$W_t(x) + W_{t-1}(y) \geq W_t(y) + W_{t-1}(x)$$

We know that $\exists a' \in x$ s.t.

$$W_t(x) = W_{t-1}(x \setminus \{a'\} \cup \{\sigma_t\}) + d(\sigma_t, a')$$

by quasiconvexity, $\exists b' \in y$ s.t.

$$W_{t-1}(x \setminus \{a'\} \cup \{\sigma_t\}) + W_{t-1}(y) \geq W_{t-1}(x \setminus \{a'\} \cup \{b'\}) + W_{t-1}(y \setminus \{b'\} \cup \{\sigma_t\})$$

as x is a minimizer,

$$W_{t-1}(x \setminus \{a'\} \cup \{b'\}) = \sum_{a \in x \setminus \{a'\} \cup \{b'\}} d(\sigma_t, a) \geq W_{t-1}(x) - \sum_{a \in x} d(\sigma_t, a)$$

$$\text{or: } W_{t-1}(x \setminus \{a'\} \cup \{b'\}) + d(\sigma_t, a') \geq W_{t-1}(x) + d(\sigma_t, b')$$

$$\begin{aligned} W_t(x) + W_{t-1}(y) &= W_{t-1}(x \setminus \{a'\} \cup \{\sigma_t\}) + d(\sigma_t, a') + \\ &+ W_{t-1}(y) \geq W_{t-1}(x \setminus \{a'\} \cup \{b'\}) + d(\sigma_t, a') + \\ &+ W_{t-1}(y \setminus \{b'\} \cup \{\sigma_t\}) \geq W_{t-1}(x) + d(\sigma_t, b') \\ &+ W_{t-1}(y \setminus \{b'\} \cup \{\sigma_t\}) \geq W_{t-1}(x) + W_t(y) \end{aligned}$$

questions:

1. Interpret the work function algorithm in the case of a uniform metric space (i.e., the paging problem).
- *2. Prove that the WFA is 2-competitive for the 2-server problem in any metric space (of any size).

randomized competitive ratio

a randomized online algorithm $\tilde{\alpha}$ is a probability distribution over deterministic online algorithms α

def: $c(\sigma, \tilde{\alpha}) \triangleq \mathbb{E}_{\alpha \sim \tilde{\alpha}} [c(\sigma, \alpha)]$

$$C_{\tilde{\alpha}} = \inf \left\{ C \mid \exists A \forall \sigma \in S_{\text{ADV}} \quad c(\sigma, \tilde{\alpha}) \leq C \cdot c_{\text{base}}(\sigma) + A \right\}$$

$$C_{\langle \text{game} \rangle}^{\text{obl}} = \inf_{\tilde{\alpha}} C_{\tilde{\alpha}}$$

distributional lower bound:

$\tilde{\sigma}$ is a probability distribution over S_{ADV}

$$L_{\tilde{\sigma}} = \sup \left\{ L \mid \forall A \forall \alpha \in S_{\text{ONL}} \quad \mathbb{E}_{\sigma \sim \tilde{\sigma}} [c(\sigma, \alpha)] \geq L \cdot \mathbb{E}_{\sigma \sim \tilde{\sigma}} [c_{\text{base}}(\sigma)] + A \right\}$$

$$L_{\langle \text{game} \rangle}^{\text{obl}} = \sup_{\tilde{\sigma}} L_{\tilde{\sigma}}$$

remark: We're ignoring the intricacies of defining σ -algebras on S_{ADV} and S_{ONL} . In examples no issues arise.

von-Neumann's minimax thm: Let $X \subset \mathbb{R}^m$ and $Y \subset \mathbb{R}^n$ be compact convex sets. If $f: X \times Y \rightarrow \mathbb{R}$ is continuous, $\forall y \in Y \quad f(\cdot, y): X \rightarrow \mathbb{R}$ is concave, and $\forall x \in X \quad f(x, \cdot): Y \rightarrow \mathbb{R}$ is convex, then

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \max_{x \in X} f(x, y)$$

(there are generalizations, e.g. Sion's minimax thm)
special case: finite two-person zero-sum game

$$X = \{x \in \mathbb{R}^m : \|x\|_1 = 1 \wedge x \geq 0\}$$

$$Y = \{y \in \mathbb{R}^n : \|y\|_1 = 1 \wedge y \geq 0\}$$

$f(x, y) = x^T A y$ where $A \in \mathbb{R}^{m \times n}$ is the utility matrix of the game

x is a probability distribution over the pure strategies of the max player

y is a probability distribution over the pure strategies of the min player

$$\max_{x \in X} \min_{y \in Y} x^T A y = \max \left\{ \mu \mid \forall j \sum_{i=1}^m a_{ij} x_i \geq \mu \wedge x \in X \right\} =$$

minimizing $y = (0, \dots, 0, 1, 0, \dots, 0)$

$$= \min \left\{ \nu \mid \forall i \sum_{j=1}^n a_{ij} y_j \leq \nu \wedge y \in Y \right\} =$$

LP duality

$$= \min_{y \in Y} \max_{x \in X} x^T A y$$

maximizing $x = (0, \dots, 0, 1, 0, \dots, 0)$

weak duality:

$$\begin{aligned} &\max \mu \\ &\text{s.t. } Ax - \begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix} \geq 0 \\ &\quad \sum_{i=1}^m x_i = 1 \\ &\quad x \geq 0 \end{aligned}$$

consider a combination $y^T A x - y^T \begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix} + \nu \sum_{i=1}^m x_i$

of the l.h.s. of the constraints.

if $y \geq 0$

$$y^T A x - y^T \begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix} + v \underbrace{\sum_{i=1}^m x_i}_{=1} \geq v$$

hence $y^T A x \geq y^T \begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix}$

we can scale $\sum_{j=1}^n y_j = 1$, so $y^T A x \geq \mu$

marking algorithm for paging:

each time a page is requested, it is marked
when $k+1$ distinct pages are marked, all
markings except for the last one are erased
serving a request: (Initially the pages in the cache are all marked)

if it's in the cache, do nothing

o.w., choose an unmarked page in the cache,
uniformly at random, and evict it

Thm: For every $k, n \in \mathbb{N}$, $k < n$, the marking
algorithm on n pages, with a cache of size k ,
is $2H_k$ -competitive, where $H_k = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} \approx \ln k$

proof: Let $\sigma = \sigma_1, \sigma_2, \dots$ be the request sequence
and let i be the first index for which σ_i is
not in the cache. W.l.o.g. we may assume
that $i=1$.

Marking implicitly divides σ into phases. A
phase that starts at σ_{j_1} ends at the smallest
 j_2 for which $\{\sigma_{j_1}, \sigma_{j_1+1}, \dots, \sigma_{j_2}, \sigma_{j_2+1}\}$ contains
 $k+1$ distinct pages.

We may assume that OPT is lazy (doesn't move
if it doesn't need to move).

In a phase, a page is clean if it was not requested in the previous phase.

It is stale if it was requested in the previous phase but not yet in the current phase.

l = #requests to clean pages in current phase

claim: OPT pays in a phase at least $\frac{1}{2}l$ (amortizes)

proof: let d be the #pages in OPT's cache but not in marking's cache in the start of the phase.

let d' be this number at the end of the phase.

$\text{cost}_{\text{OPT}}(\text{phase}) \geq l - d$, because none of the l clean pages requested are in marking's cache, and at most d of those are in OPT's cache.

Now, let S denote the set of marked pages at the end of the phase. Exactly d' of those are not in OPT's cache at the end of the phase. Thus, at least d' cache slots of OPT never contained a page from S . So $k - d'$ slots were used to serve all requests in S ($|S| = k$), implying that OPT had to evict at least d' pages.

thus $\text{cost}_{\text{OPT}}(\text{phase}) \geq \max\{l - d, d'\} \geq \frac{1}{2}(l - d + d')$

summing over phases, the d, d' terms telescope.

claim: The expected cost of marking in the phase is at most $l H_k$.

proof: Each of the l clean pages requested costs 1. There are $k - l$ stale pages requested.

The probability that the first request to a stale page cost 1 is $\frac{c}{s}$ where c is the number of clean pages requested so far

and s is the current number of stale pages.
The worst case is when all l clean pages are requested first, increasing c .

$$E[\text{cost}] \leq \frac{l}{k} + \frac{l}{k-1} + \frac{l}{k-2} + \dots + \frac{l}{l+1} = l(H_k - H_l)$$

Thus, the total cost is $l(H_k - H_l) + l \leq l H_k$.

lower bound

The idea is to construct a probability distribution $\tilde{\sigma}$ over request sequences.

Claim: If $\forall n \exists$ prob. distrib. $\tilde{\sigma}_n$ on requests of length n s.t.

$$(1) \lim_{n \rightarrow \infty} E[C_{\text{base}}(\sigma) : \sigma \sim \tilde{\sigma}_n] = \infty$$

$$(2) \forall \epsilon > 0 \exists n_0 \forall n \geq n_0 \forall \text{det. online alg } A, \\ E[C_A(\sigma) : \sigma \sim \tilde{\sigma}_n] > (c - \epsilon) E[C_{\text{base}}(\sigma) : \sigma \sim \tilde{\sigma}_n]$$

then the randomized competitive ratio is $\geq c$

proof: Assume for contradiction that there exists a randomized online alg. $\tilde{A}$ which is c' -competitive for $c' < c$.

In particular, this implies that $\exists$ const. s.t.

$$\forall \sigma, E[C_{\tilde{A}}(\sigma) : A \sim \tilde{A}] \leq c' \cdot C_{\text{base}}(\sigma) + \text{const.}$$

$$\text{So, } E[E[C_{\tilde{A}}(\sigma) : A \sim \tilde{A}] : \sigma \sim \tilde{\sigma}_n] \leq c' E[C_{\text{base}}(\sigma) : \sigma \sim \tilde{\sigma}_n] + \text{const.}$$

Pick $\epsilon \leq \frac{c - c'}{2}$ and $n \geq n_0$ sufficiently large so that $\text{const.} \leq \epsilon \cdot E[C_{\text{base}}(\sigma) : \sigma \sim \tilde{\sigma}_n]$.

$\exists A$ in the support of $\tilde{A}$ (which must be finite

for a sequence of length n) s.t.

$$\begin{aligned} E[C_A(\sigma) : \sigma \sim \tilde{\sigma}_n] &\leq E[E[C_A(\sigma) : A \sim \tilde{A}] : \sigma \sim \tilde{\sigma}_n] \leq \\ &\leq c' E[C_{\text{base}}(\sigma) : \sigma \sim \tilde{\sigma}_n] + \text{const.} \leq (c' + \epsilon) \cdot E[C_{\text{base}}(\sigma) : \sigma \sim \tilde{\sigma}_n] \\ &\leq (c' - \epsilon) \cdot E[C_{\text{base}}(\sigma) : \sigma \sim \tilde{\sigma}_n], \text{ a contradiction.} \end{aligned}$$

Thm: The randomized competitive ratio of paging is at least $H_k = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} \sim \ln k$.

proof: We'll restrict attention to $k+1$ pages.

for n , $\tilde{\sigma}$ is simply choosing n independent pages, each uniformly distributed among all $k+1$ pages. Any online alg. pays $\frac{n}{k+1}$ in expectation.

To estimate OPT, we partition the sequence into phases. Phase 1 begins with the first request. a new phase begins when $k+1$ distinct pages have been requested since the start of the previous phase. (The last page requested starts the new phase)
Let X_i denote the number of requests needed to raise the number of pages requested in a phase so far from i to $i+1$

X_i is a geometric random variable with success probability $\frac{k+1-i}{k+1}$ so $E[X_i] = \frac{k+1}{k+1-i}$

$$\begin{aligned} \text{Therefore } E[\text{length of a phase}] &= \sum_{i=1}^k \frac{k+1}{k+1-i} = \\ &= (k+1) \left(\frac{1}{k} + \frac{1}{k-1} + \dots + 1 \right) = \\ &= (k+1) H_k \end{aligned}$$

Let F_n denote the number of complete phases in n requests. By the elementary renewal theorem, $\lim_{n \rightarrow \infty} \frac{E[F_n]}{n} = \frac{1}{(k+1)H_k}$

OPT pays at most F_n (because it knows the request that will start the next phase).

Thus, for every $\epsilon > 0 \exists n_0$ s.t. $\forall n \geq n_0$.

$$E[F_n] < \frac{n}{(k+1)H_k} (1+\epsilon)$$

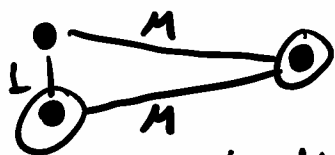
$$\text{so, } \forall A, E[C_A(\tilde{\sigma})] = \frac{n}{k+1} > \frac{H_k}{1+\epsilon} E[F_n] \geq$$

$$\geq \left(H_k - \frac{\epsilon H_k}{1-\epsilon} \right) E[C_{\text{base}}(\tilde{\sigma})].$$

the n-step distrib.

questions:

1. Consider the following 3-point metric space:



The circles mark the initial server positions (2 servers).

Show that the randomized competitive ratio, for some value of M , is $> H_2$.

2. Consider the following randomized list update algorithm: for each element x in the list choose $b_x \in \{0, 1\}$ independently and uniformly at random. When x is requested, flip b_x and then move x to front iff $b_x = 1$ (o.w., don't move x at all).

Show that this alg. is better than 2-comp.

Weighted caching

Similar to paging: we have n pages $\{1, \dots, n\}$ k of them fit into the cache. Loading page i into the cache costs p_i (paging is when all $p_i = 1$).

Denote:

$$x_{ij} = \begin{cases} 1 & \text{if page } i \text{ removed from cache between } j\text{-th} \\ & \text{and } j+1\text{st request to this page (or the end of the seq.)} \\ 0 & \text{o.w.} \end{cases}$$

t_{ij} = the time page i is requested for the j -th time

n_{it} = the number of requests to page i until time t

$$B(t) = \{i : n_{it} > 0\}$$

i_t = the request at time t

note that after serving the time t request, i_t is in the cache. of the remaining $|B(t)| - 1$ pages we've seen so far, at most $k - 1$ are in the cache.

So, we must have removed at least $|B(t)| - 1 - (k - 1) = |B(t)| - k$ of those pages.

We can express OPT as follows (T requests):

$$\min \sum_{i=1}^n \sum_{j=1}^{n_{iT}} p_i x_{ij} \text{ subject to:}$$

we charge for removing i rather than loading i

$$\sum_{i \in B(t) \setminus \{i_t\}} x_{i, n_{it}} \geq |B(t)| - k$$

$$x_{ij} \in \{0, 1\}$$

$$\forall t = 1, 2, \dots, T$$

$$\forall i \in \{1, \dots, n\}$$

$$\forall j \in \{1, \dots, n_{iT}\}$$

if we relax $x_{ij} \in \{0, 1\}$ to $x_{ij} \in [0, 1]$ we get a lower bound on OPT .

the dual program is:

$$\max \sum_{t=1}^T (B(t) - k) y_t - \sum_{i=1}^n \sum_{j=1}^{n_{i,T}} z_{ij}$$

subject to

$$\sum_{t=t_{ij}+1}^{t_{i,j+1}} y_t - z_{ij} \leq p_i$$

$$\forall i \in \{1, \dots, n\}$$

$$\forall j \in \{1, \dots, n_{i,T}\}$$

$$\forall t \in \{1, \dots, T\}$$

$$\forall i, j$$

$$y_t \geq 0$$

$$z_{ij} \geq 0$$

online algorithm:

We don't know the primal or the dual program in advance. Each request exposes a new primal constraint and a new dual variable.

We begin with all (primal and dual) variables = 0 at time t , we start raising y_t (think of this as a continuous process).

for all $x_{i,t} < 1$ maintain:

$$\frac{dx_{i,t}}{dy_t} = \frac{\ln(1+k)}{p_i} \left(x_{i,t} + \frac{1}{k} \right) \quad (*)$$

for all $x_{i,t} = 1$ we raise $z_{i,t}$ at the same rate as y_t

we stop when the new primal constraint becomes tight.

Note that by construction, at the end x is a feasible primal solution.

what about the dual solution? -

Note that
$$x_{ij} = \frac{1}{k} \left(e^{\frac{\ln(1+k)}{P_i} \left(\sum_{t=t_{ij}+1}^{t_{i(j+1)}} y_t - \beta_{ij} \right)} - 1 \right)$$

So, $x_{ij} \leq 1 \Rightarrow \frac{\ln(1+k)}{P_i} \left(\sum_{t=t_{ij}+1}^{t_{i(j+1)}} y_t - \beta_{ij} \right) \leq \ln(1+k)$

$$\Rightarrow \sum_{t=t_{ij}+1}^{t_{i(j+1)}} y_t - \beta_{ij} \leq P_i$$

so, (y, β) is a feasible dual solution.

Now, consider increasing y_t at a rate $\frac{dy_t}{d\tau} = 1$

and let $S_{t,\tau}$ denote the set of pages for which $x_{init} < 1$ at τ .

$$\frac{d \sum_{ij} P_i x_{ij}}{d\tau} = \frac{d \sum_{ij} P_i x_{ij}}{dy_t} \cdot \underbrace{\frac{dy_t}{d\tau}}_{=1} = \sum_{i \in B(t) \setminus \{i_t\}} P_i \frac{dx_{init}}{dy_t} =$$

$$= \ln(1+k) \sum_{i \in S_{t,\tau} \setminus \{i_t\}} \left(x_{init} + \frac{1}{k} \right) =$$

$$= \ln(1+k) \left(\sum_{i \in B(t) \setminus \{i_t\}} x_{init} - |B(t) \setminus S_{t,\tau}| + \frac{|S_{t,\tau}| - 1}{k} \right) \leq$$

$$\leq \ln(1+k) \left(|B(t)| - k - |B(t) \setminus S_{t,T}| + \frac{|S_{t,T}| - 1}{k} \right) \leq$$

$\uparrow$
 because the
 primal constraints
 are never violated

$|S_{t,T}| - k + \frac{|S_{t,T}| - 1}{k}$

$$\leq 2 \ln(1+k) (|S_{t,T}| - k) =$$

$$= 2 \ln(1+k) \frac{\sum_{t=1}^T (|B(t)| - k) y_t - \sum_{i=1}^n \sum_{j=1}^{n_{i,T}} z_{ij}}{\sum y_t}$$

Thus, at the end,

$$\sum_{ij} p_{ij} x_{ij} \leq 2(1 + \ln k) \left(\sum_t (|B(t)| - k) y_t - \sum_{ij} z_{ij} \right)$$

$$\leq 2(1 + \ln k) \text{OPT}$$

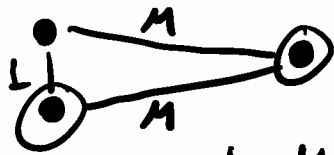
$\nearrow$
 weak duality

to recap: we showed how to compute online a fractional primal solution x that $2(1 + \ln k)$ -approximates OPT .

We'll see next week how to convert a fractional solution to a randomized algorithm.

questions:

1. Consider the following 3-point metric space:



The circles mark the initial server positions (2 servers).

Show that the randomized competitive ratio, for some value of M , is $> H_2$.

2. Show that if we set in (*)

$$\frac{dx_{init}}{dy_t} = \frac{\ln(1+k \ln k)}{P_i} \left(x_{init} + \frac{1}{k \ln k} \right)$$

then we get a competitive ratio $(1+o(1)) \ln(1+k)$.

example: $n=4$. pages 1, 2 have weight 1, pages 3, 4 have weight $M \gg 1$. $k=2$.

suppose we move from fractional state $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ to fractional state $(0, 1, \frac{1}{2}, \frac{1}{2})$. the cheapest way to do this is to evict $\frac{1}{2}$ of page 1 and load $\frac{1}{2}$ of page 2, a cost of $\frac{1}{2}$.

However, if $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ represents the state of a randomized algorithm which is a cache of $\{1, 2\}$ with prob. $\frac{1}{2}$ and a cache of $\{3, 4\}$ with prob. $\frac{1}{2}$, there is no way to get $(0, 1, \frac{1}{2}, \frac{1}{2})$ by paying less than $\frac{M}{2}$. For instance, if $(0, 1, \frac{1}{2}, \frac{1}{2})$ represents $\{2, 3\}$ w/prob. $\frac{1}{2}$ and $\{2, 4\}$ w/prob. $\frac{1}{2}$, the switch costs $\frac{M+1}{2}$ (load 3 instead of 1 to $\{1, 2\}$ and load 2 instead of 3 to $\{3, 4\}$).

So, the problem is non-trivial.

To solve it, we scale weight so that the minimum weight is 1, and we round them (up) to the nearest power of 2. In each power of 2 class let's number the pages separately.

We'll define a cost of 2^{i-1} for both loading and evicting a page of weight 2^i .

Let $q_{i,j}(t) \in [0, 1]$ denote the fraction of the j -th page of weight 2^i that is in the cache at time t . So, $\sum_{i,j} q_{i,j}(t) = k$ for all t .

Now, set $k_i(t) = \sum_j q_{ij}(t)$, the fractional cache volume that weight 2^i pages take.

We will interpret a fractional cache as a probability distribution over integral caches as follows:

pick α uniformly at random in $[0,1)$. This determines a cache $C_\alpha(t)$ as follows:

Set $T_\alpha = \{\alpha, 1+\alpha, 2+\alpha, \dots, k-1+\alpha\}$

put in $C_\alpha(t)$ $n_i(t)$ weight 2^i pages, for all i ,

where $n_i(t) = |T_\alpha \cap [k_0(t) + k_1(t) + \dots + k_{i-1}(t), k_0(t) + \dots + k_i(t))|$

note that $n_i(t) \in \{\lfloor k_i(t) \rfloor, \lceil k_i(t) \rceil\}$.

Which pages are chosen? - for class i we partition $[k_0(t) + \dots + k_{i-1}(t), k_0(t) + \dots + k_i(t))$ (an interval of length $k_i(t)$) into subintervals (shifted here by $k_0(t) + \dots + k_{i-1}(t)$ to simplify the notation) $[0, q_{i1}), [q_{i1}, q_{i1} + q_{i2}), \dots$

Then put in the cache the pages corresponding to subintervals "hit" by T_α .

How do we update the distribution? -

The fractional cost is just a sequence of transfers of ε mass from page i, j to page i', j' .

case 1: $i = i'$. We evict i, j from ε fraction of the caches that contain it and we load i, j' to ε fraction of the caches that don't contain it.

Now, every cache from which i, j was evicted

has $\lfloor k_i(t) \rfloor$ or $\lfloor k_i(t) \rfloor - 1$ weight 2^i pages,
 and every cache to which i, j was loaded
 has $\lfloor k_i(t) \rfloor$ or $\lfloor k_i(t) \rfloor + 1$ weight 2^i pages.
 We match (fractionally) caches with
 deficit to caches with excess, and move
 one weight 2^i page between matched
 caches (from excess to deficit).
Total cost: at most $2 \cdot 2 \cdot \epsilon \cdot 2^{i-1} = 2\epsilon 2^i$,
 twice the fractional cost.

case 2: $i < i'$ (the other direction is
 analogous).

In this case: $k_0 + k_1 + \dots + k_i$ moves ϵ
 to the left, the interval for classes
 $i+1, i+2, \dots, i'-1$ move ϵ to the left (but
 don't change length) and the left
 boundary $k_0 + k_1 + \dots + k_{i'-1}$ of the interval
 for i' moves ϵ to the left.

for $i < h < i'$, ϵ fraction of the caches
 (with $\lfloor k_h(t) \rfloor$ weight h pages) lose a weight
 h page, and ϵ fraction (with $\lfloor k_h(t) \rfloor$ such
 pages) gain a weight h page. We match
 losers and gainers and move a page
 between each pair. The total cost of
 this adjustment is $\sum_{h=i+1}^{i'-1} \epsilon \cdot 2 \cdot 2^{h-1} = \epsilon \sum_{h=i+1}^{i'-1} 2^h$.

Next we remove the page i, j from
 the ϵ fraction of mass of the caches
 that should no longer contain it. Each
 such cache has now $\lfloor k_i \rfloor$ or $\lfloor k_i \rfloor - 1$
 weight i pages. We match caches in
 deficit to caches with $\lfloor k_i \rfloor$ such

pages and move a page to cover the deficit.

We do an analogous correction with i', j' .

The total cost is at most

$$\underbrace{\epsilon \cdot 3 \cdot 2^{i-1}}_{\text{for fixing } i, j} + \underbrace{\epsilon \cdot 3 \cdot 2^{i'-1}}_{\text{for fixing } i', j'} + \underbrace{\epsilon \sum_{h=i+1}^{i'-1} 2^h}_{\text{for the rest}} \leq$$

$$\leq 3\epsilon 2^{i-1} + 5\epsilon 2^{i'-1} \leq 5(\underbrace{\epsilon(2^{i-1} + 2^{i'-1})}_{\text{fractional cost}})$$

So, converting the fractional solution online to a randomized algorithm costs an extra factor of 5.

questions:

1. Ski rental is a trivial online problem:

At each step, the adversary decides: continue or stop. Stop terminates the game. If it continues, the online algorithm decides: rent or buy. Rent costs 1. Buy costs $M \gg 1$, but following buy the algorithm does not need to pay for future steps.

Design a primal-dual randomized online ski rental algorithm with better than 2 competitive ratio.

2. In layered graph traversal, a layered graph $G = (V_0, V_1, V_2, \dots, E)$ endowed with non-negative edge lengths $\ell: E \rightarrow \mathbb{N}$ (including 0) is revealed layer by layer. A searcher begins at V_0 (wlog $V_0 = \{s\}$)

and can move forward and backward along the edges, paying the total length traveled. When the searcher reaches a node in V_i , all the layer V_{i+1} and all the edges between V_i and V_{i+1} are revealed. The goal is to reach a target node t in the last layer (which wlog contains just t).

- a. Show that metrical task systems are a special case of layered graph traversal.
- b. Define a fractional solution to layered graph traversal and show that it can be converted on-the-fly into a randomized algorithm at no loss of cost.

a quick explanation of online mirror descent applied to paging

we have n pages and a cache of size k

the state space is $\{x \in \{0,1\}^n : \sum_i x_i = n-k\}$, where
 $x_i = 1$ iff page i is not in the cache.

fraction solutions can use states in $\{x \in [0,1]^n : \sum_i x_i = n-k\}$
we saw in a previous lecture how to convert a fractional solution into a randomized algorithm.

mirror descent: to serve a request at i follow
the dynamic $\nabla^2 \phi(x(t)) \dot{x}(t) \in -e_i - \lambda(x(t))$ until
 x_i reaches $\frac{1}{2k}$ (instead of the obvious 0).

Here $\phi(x) = \sum x_i \ln x_i$ is the entropic
regularizer, $e_i \in \mathbb{R}^n$ is the vector with 1
in the i th coordinate and 0 elsewhere (the
control function driving x_i down), and
 $\lambda(x)$ is a vector in the normal cone to
the feasible region, at x (the Lagrange
multipliers that prevent movement outside
the feasible region).

$$\nabla \phi(x) = (1 + \ln x_i)_{i=1}^n \quad \nabla^2 \phi(x) = \begin{pmatrix} \ddots & & 0 \\ & \frac{1}{x_i} & \\ 0 & & \ddots \end{pmatrix}$$

ϕ is a strongly convex function. The Bregman
divergence associated with ϕ is:

$$D_\phi(y \| x) = \phi(y) - \phi(x) - \langle \nabla \phi(x), y - x \rangle$$

We'll use $\hat{D}_\phi(y||x) = -\phi(x) - \langle \nabla \phi(x), y-x \rangle$ as a potential function, where y is the adversary state and x is the online algorithm's state.

ADV move: *fixed*

Hölder Inequality

$$\frac{\partial}{\partial t} \hat{D}_\phi(y(t)||x) = -\langle \nabla \phi(x), \dot{y}(t) \rangle \leq$$

$$\leq \|\nabla \phi(x)\|_\infty \|\dot{y}(t)\|_1 \leq \|\dot{y}(t)\|_1 \ln 2k$$

ADV movement cost (up to a factor of 2) *comp. ratio will be $O(\ln k)$*

ALG movement: *fixed, $y_i=0$*

$$\begin{aligned} \frac{\partial}{\partial t} \hat{D}_\phi(y||x(t)) &= -\langle \nabla \phi(x(t)), \dot{x}(t) \rangle + \\ &+ \langle \nabla \phi(x(t)), \dot{x}(t) \rangle - \langle \dot{x}(t) | \nabla^2 \phi(x(t)) | y - x(t) \rangle \\ &= \langle -e_i - \lambda(x(t)), y - x(t) \rangle \end{aligned}$$

$$= \langle e_i + \lambda(x(t)), y - x(t) \rangle \leq$$

$$\leq y_i - x_i(t) = -x_i(t)$$

at least ALG's cost as:

as $\lambda \in$ normal cone @ x and y is in the feasible region

$$\langle \lambda, y - x \rangle \leq 0$$

$$\dot{x}_i(t) = -x_i(t) - \lambda_i(x(t))$$

and the projection λ would reduce the movement

$l_{i,t}$ = loss of arm i at step $t \in [0, T]$

the algorithm is randomized & the adversary is adaptive, hence the $l_{i,t}$ -s are random variables

(gain: $g_{i,t} = 1 - l_{i,t}$)

i_t - the arm that the algorithm chooses at step t
(also a random variable)

the regret: $R_T = \sum_{t=1}^T l_{i_t, t} - \min_{i \in \{1, \dots, n\}} \sum_{t=1}^T l_{i, t}$

We're interested in $E[R_T]$

pseudo-regret:

$$\bar{R}_T = E\left[\sum_{t=1}^T l_{i_t, t}\right] - \min_{i \in \{1, \dots, n\}} E\left[\sum_{t=1}^T l_{i, t}\right] \leq E[R_T]$$

(if the adversary is oblivious, =)

EXP3:

$$\forall i \in \{1, \dots, n\}, \tilde{L}_{i,0} \leftarrow 0$$

$P_1 \leftarrow$ uniform distribution on $\{1, \dots, n\}$

for $t \leftarrow 1, 2, \dots, T$:

draw i_t from P_t $l_{i_t, t} / P_{i_t, t}$ $i = i_t$

$$\forall i \in \{1, \dots, n\}, \tilde{L}_{i, t} \leftarrow \begin{cases} l_{i, t} / P_{i, t} \\ 0 \end{cases} \quad \text{o.w.}$$

$$\tilde{L}_{i, t} \leftarrow \tilde{L}_{i, t-1} + \tilde{L}_{i, t}$$

$$\forall i \in \{1, \dots, n\}, P_{i, t+1} \leftarrow \frac{e^{-\eta_t \tilde{L}_{i, t}}}{\sum_{j=1}^n e^{-\eta_t \tilde{L}_{j, t}}}$$

notice that $\tilde{l}_{i,t}$ is an unbiased estimator of $l_{i,t}$:

$$E_{i_t \sim p_t}[\tilde{l}_{i,t}] = \sum_{j=1}^n \Pr[i_t=j] \cdot \chi_{i=j} \frac{l_{i,t}}{p_{i,t}} = l_{i,t}$$

the η_t -s are a non-increasing sequence

Claim: For every non-increasing $(\eta_t)_{t \in \mathbb{N}}$,
EXP3 satisfies

$$\bar{R}_T \leq \frac{n}{2} \sum_{t=1}^T \eta_t + \frac{\ln n}{\eta_T}$$

Corollary:

(i) For $\eta_t = \eta = \sqrt{\frac{2 \ln n}{n T}}$, $\bar{R}_T \leq \sqrt{2 T n \ln n}$

(ii) For $\eta_t = \sqrt{\frac{\ln n}{n t}}$, $\bar{R}_T \leq 2 \sqrt{T n \ln n}$

proof of claim:

Note: $E_{i_t \sim p_t}[\tilde{l}_{i,t}] = l_{i_t,t}$ (r.v.)

$$E_{i_t \sim p_t}[\tilde{l}_{i,t}] = l_{i_t,t}$$

$$E_{i_t \sim p_t}[\tilde{l}_{i,t}^2] = \frac{l_{i_t,t}^2}{p_{i_t,t}} \quad (\text{r.v.})$$

$$E_{i_t \sim p_t}\left[\frac{1}{p_{i_t,t}}\right] = n$$

$$\forall j, \quad \sum_{t=1}^T l_{i_t, t} - \sum_{t=1}^T l_{j, t} = \sum_{t=1}^T E_{i \sim p_t} [\tilde{l}_{i, t}] - \sum_{t=1}^T E_{i \sim p_t} [\tilde{l}_{j, t}]$$

$$E_{i \sim p_t} [\tilde{l}_{i, t}] = \frac{1}{\eta_t} \ln E_{i \sim p_t} [e^{-\eta_t (\tilde{l}_{i, t} - E_{j \sim p_t} [\tilde{l}_{j, t}])}]$$

$$- \frac{1}{\eta_t} \ln E_{i \sim p_t} [e^{-\eta_t \tilde{l}_{i, t}}]$$

Now, $\forall x > 0$, $\ln x \leq x - 1$ and $e^{-x} \leq 1 - x + \frac{x^2}{2}$

$$\begin{aligned} &= \ln E_{i \sim p_t} [e^{-\eta_t \tilde{l}_{i, t}}] + \eta_t E_{j \sim p_t} [\tilde{l}_{j, t}] \\ &\leq E_{i \sim p_t} [e^{-\eta_t \tilde{l}_{i, t}}] - 1 + \eta_t \tilde{l}_{i, t} \\ &= E_{i \sim p_t} [e^{-\eta_t \tilde{l}_{i, t}} - 1 + \eta_t \tilde{l}_{i, t}] \\ &\leq E_{i \sim p_t} \left[\frac{\eta_t^2 \tilde{l}_{i, t}^2}{2} \right] \\ &\leq \frac{\eta_t^2}{2 p_{i_t, t}} \end{aligned}$$

$\tilde{l}_{i_t, t}^2 \leq 1 \rightarrow$

to bound $\quad$ define:

$$\phi_0(\eta) = 0, \quad \phi_t(\eta) = \frac{1}{\eta} \ln \frac{1}{n} \sum_{i=1}^n e^{-\eta \tilde{l}_{i, t}}$$

$$\begin{aligned}
 & -\frac{1}{\eta_t} \ln E_{i \sim p_t} [e^{-\eta_t \tilde{L}_{i,t}}] = \\
 & = -\frac{1}{\eta_t} \ln \sum_{i=1}^n \frac{e^{-\eta_{t-1} \tilde{L}_{i,t-1}} e^{-\eta_t \tilde{L}_{i,t}}}{\sum_{j=1}^n e^{-\eta_{t-1} \tilde{L}_{j,t-1}}}
 \end{aligned}$$

$$\eta_t \leq \eta_{t-1} \rightarrow \leq -\frac{1}{\eta_t} \left[\ln \sum_{i=1}^n e^{-\eta_t \tilde{L}_{i,t}} - \ln \sum_{i=1}^n e^{-\eta_{t-1} \tilde{L}_{i,t-1}} \right]$$

$$= \phi_{t-1}(\eta_{t-1}) - \phi_t(\eta_t)$$

$$\text{so, } \forall j \quad \sum_{t=1}^T l_{i_t,t} - \sum_{t=1}^T l_{j,t} \leq$$

$$\leq \sum_{t=1}^T \frac{\eta_t}{2\rho_{i_t,t}} + \sum_{t=1}^T [\phi_{t-1}(\eta_{t-1}) - \phi_t(\eta_t)] - \sum_{t=1}^T E_{i_t \sim p_t} [\tilde{L}_{j,t}]$$

$$E \left[\sum_{t=1}^T \frac{\eta_t}{2\rho_{i_t,t}} \right] = E \left[\sum_{t=1}^T E_{i_t \sim p_t} \left[\frac{\eta_t}{2\rho_{i_t,t}} \right] \right] = \frac{n}{2} \sum_{t=1}^T \eta_t$$

$$\phi_0(\eta_0) = 0$$

$$\begin{aligned}
 -\phi_T(\eta_T) &= \frac{\ln n}{\eta_T} - \frac{1}{\eta_T} \ln \sum_{i=1}^n e^{-\eta_T \tilde{L}_{i,T}} \leq \\
 &\leq \frac{\ln n}{\eta_T} - \frac{1}{\eta_T} \ln e^{-\eta_T \tilde{L}_{j,T}}
 \end{aligned}$$

$$= \frac{\ln n}{\eta_T} + \tilde{L}_{j,T}$$

$$= \frac{\ln n}{\eta_T} + \sum_{t=1}^T \tilde{\ell}_{j,t}$$

$$\begin{aligned} E \left[\sum_{t=1}^T \ell_{i_t,t} - \sum_{t=1}^T \ell_{j,t} \right] &\leq \\ &\leq \frac{\eta}{2} \sum_{t=1}^T \eta_t + \frac{\ln n}{\eta_T} + E \left[\sum_{t=1}^T E_{i_t \sim p_t} [\ell_{i_t,t}] \right] - E \left[\sum_{t=1}^T E_{j \sim p_t} [\ell_{j,t}] \right] \end{aligned}$$

In order to bound the regret, rather than the pseudo-regret, we need high probability bounds. Unfortunately, $\tilde{\ell}_{i,t}$ has variance $\approx \frac{1}{p_{i,t}}$ which is unbounded.

(partly, the problem is that we don't do enough exploration)

Let $\beta \leq 1$. Define

$$\tilde{g}_{i,t} = \begin{cases} \frac{g_{i,t} + \beta}{p_{i,t}} & i = i_t \\ \frac{\beta}{p_{i,t}} & \text{o.w.} \end{cases}$$

$$\forall \delta \in (0, 1), \text{ w.p. } \geq 1 - \delta, \sum_{t=1}^T g_{i,t} \leq \sum_{t=1}^T \tilde{g}_{i,t} + \frac{\ln \delta^{-1}}{\beta}$$

proof:

$$\begin{aligned} E[e^{\beta g_{i,t} - \beta \tilde{g}_{i,t}} | i_1, i_2, \dots, i_{t-1}] &\leq \overset{e^x \leq 1 + x + x^2 \quad \forall x \leq 1}{\left(1 + E\left[\beta g_{i,t} - \beta \tilde{g}_{i,t} - \frac{\beta^2}{P_{i,t}}\right] + E\left[\left(\beta g_{i,t} - \beta \tilde{g}_{i,t} - \frac{\beta^2}{P_{i,t}}\right)^2\right]\right)} e^{-\frac{\beta^2}{P_{i,t}}} \\ &\leq \left(1 + \beta^2 \frac{g_{i,t}^2}{P_{i,t}}\right) e^{-\frac{\beta^2}{P_{i,t}}} \leq 1 \quad \leftarrow 1+x \leq e^x \end{aligned}$$

$$\Rightarrow E\left[e^{\beta \sum_{t=1}^T (g_{i,t} - \tilde{g}_{i,t})}\right] \leq 1$$

$$\begin{aligned} \Pr\left[\beta \sum_{t=1}^T (g_{i,t} - \tilde{g}_{i,t}) > \ln \delta^{-1}\right] &= \\ &= \Pr\left[e^{\beta \sum_{t=1}^T (g_{i,t} - \tilde{g}_{i,t})} > \frac{1}{\delta}\right] \\ &\leq \delta E\left[e^{\beta \sum_{t=1}^T (g_{i,t} - \tilde{g}_{i,t})}\right] \leq \delta \end{aligned}$$

Markov's ineq.

EXP3.P $\eta \in (0, \infty)$ $\beta, \delta \in [0, 1]$

$P_t \leftarrow$ uniform distrib. on $\{1, \dots, n\}$

For $t \leftarrow 1, 2, \dots, T$:

draw i_t from P_t

$$V_i, \tilde{g}_{i,t} \leftarrow \begin{cases} (g_{i,t} + \beta) / P_{i,t} & i = i_t \\ \beta / P_{i,t} & \text{o.w.} \end{cases}$$

$$\tilde{G}_{i,t} \leftarrow \sum_{s=1}^t \tilde{g}_{i,s}$$

compute:

$$V_i, P_{i,t+1} \leftarrow (1-\delta) \frac{e^{\eta \tilde{G}_{i,t}}}{\sum_{j=1}^n e^{\eta \tilde{G}_{j,t}}} + \frac{\delta}{n}$$

Thm: $\forall \delta \in (0, 1)$, if $\eta = 0.95 \sqrt{\frac{\ln n}{nT}}$, $\beta = \sqrt{\frac{\ln(\delta^{-1}n)}{nT}}$,

$\delta = 1.05 \sqrt{\frac{n \ln n}{T}}$, then w.p. $\geq 1 - \delta$

$$R_T \leq 5.15 \sqrt{T n \ln(n\delta^{-1})}$$

If we replace $\beta = \sqrt{\frac{\ln n}{nT}}$ then w.p. $\geq 1 - \delta$

$$R_T \leq \sqrt{\frac{nT}{\ln n}} \ln \delta^{-1} + 5.15 \sqrt{T n \ln n}$$

proof: $E_{i \sim P_t}[\tilde{g}_{i,t}] = g_{i_t,t} + \beta n$

Therefore

$$V_i, \sum_{t=1}^T g_{j,t} - \sum_{t=1}^T g_{i_t,t} = \beta n T + \sum_{t=1}^T g_{j,t} - \sum_{t=1}^T E_{i \sim P_t}[\tilde{g}_{i,t}]$$

define $u = \text{uniform distribution on } \{1, \dots, n\}$

$$q_t = \frac{1}{1-\delta} (p_t - \delta u) \quad (\text{i.e. distrib. w/out the mixing})$$

$$-E_{i \sim p_t}[\tilde{g}_{i,t}] = -(1-\delta) E_{i \sim q_t}[\tilde{g}_{i,t}] - \delta E_{i \sim u}[\tilde{g}_{i,t}]$$

$$= (1-\delta) \left(\frac{1}{\eta} \ln E_{i \sim q_t} [e^{\eta(\tilde{g}_{i,t} - E_{i \sim q_t}[\tilde{g}_{i,t}])}] - \right.$$

$$\left. - \frac{1}{\eta} \ln E_{i \sim q_t} [e^{\eta \tilde{g}_{i,t}}] \right) - \delta E_{i \sim u}[\tilde{g}_{i,t}]$$

let's assume $r \leq \frac{1}{2}$ and $(1+\beta)\eta n \leq \delta$

$$\ln E_{i \sim q_t} [e^{\eta(\tilde{g}_{i,t} - E_{i \sim q_t}[\tilde{g}_{i,t}])}] =$$

$$= \ln E_{i \sim q_t} [e^{\eta \tilde{g}_{i,t}}] - \eta E_{i \sim q_t}[\tilde{g}_{i,t}] \leq$$

$$\leq E_{i \sim q_t} [e^{\eta \tilde{g}_{i,t}} - 1 - \eta \tilde{g}_{i,t}] \leq$$

$$\leq E_{i \sim q_t} [\eta^2 \tilde{g}_{i,t}^2] = \eta^2 \sum_{i=1}^n q_{i,t} \tilde{g}_{i,t}^2 = \eta^2 \sum_{i=1}^n \frac{q_{i,t}}{p_{i,t}} (q_{i,t} + \beta) \tilde{g}_{i,t} \leq$$

$$\frac{q_{i,t}}{p_{i,t}} \leq \frac{1}{1-\delta} \rightarrow \frac{1+\beta}{1-\delta} \eta^2 \sum_{i=1}^n \tilde{g}_{i,t}$$

$$\text{Now, } q_{i,t} = \frac{e^{\eta \tilde{g}_{i,t-1}}}{\sum_{j=1}^n e^{\eta \tilde{g}_{j,t-1}}}$$

where $\forall i, \tilde{g}_{i,0} = 0$

So,

$$-\sum_{t=1}^T E_{i \sim p_t} [\tilde{g}_{i,t}] \leq$$

$$\leq \sum_{t=1}^T \left[(1-\delta) \left(\frac{1}{\eta} \cdot \frac{1+\beta}{1-\delta} \eta^2 \sum_{i=1}^n \tilde{g}_{i,t} - \frac{1}{\eta} \ln \sum_{i=1}^n g_{i,t} e^{\eta \tilde{g}_{i,t}} \right) - \delta \sum_{i=1}^n \frac{1}{n} \tilde{g}_{i,t} \right] \leq$$

$$\leq (1+\beta) \eta \sum_{t=1}^T \sum_{i=1}^n \tilde{g}_{i,t} - \frac{1-\delta}{\eta} \sum_{t=1}^T \ln \sum_{i=1}^n \frac{e^{\eta \tilde{G}_{i,t-1}}}{\sum_{i=1}^n e^{\eta \tilde{G}_{i,t-1}}} e^{\eta \tilde{g}_{i,t}}$$

$$= (1+\beta) \eta \sum_{t=1}^T \sum_{i=1}^n \tilde{g}_{i,t} - \frac{1-\delta}{\eta} \ln \prod_{t=1}^T \frac{\sum_{i=1}^n e^{\eta \tilde{G}_{i,t}}}{\sum_{i=1}^n e^{\eta \tilde{G}_{i,t-1}}} \leq$$

$$\leq (1+\beta) \eta n \max_j \tilde{G}_{j,T} - \frac{1-\delta}{\eta} \ln \frac{\sum_{i=1}^n e^{\eta \tilde{G}_{i,T}}}{\sum_{i=1}^n e^{\eta \tilde{G}_{i,0}}} =$$

$$= (1+\beta) \eta n \max_j \tilde{G}_{j,T} + \frac{1-\delta}{\eta} \ln n -$$

$$- \frac{1-\delta}{\eta} \ln \sum_{i=1}^n e^{\eta \tilde{G}_{i,T}} \leq$$

$$\underbrace{\geq e^{\eta \max_j \tilde{G}_{j,T}}}_{\eta \max_j \tilde{G}_{j,T}}$$

$$\leq -(1-\delta - (1+\beta) \eta n) \cdot \max_j \tilde{G}_{j,T} + \frac{\ln n}{\eta} \leq$$

$$\leq -(1-\delta - (1+\beta) \eta n) \max_j \sum_{t=1}^T \tilde{g}_{i,t} + \frac{\ln(\delta^{-1} n)}{\beta} + \frac{1}{1-\delta} \frac{\ln n}{n}$$

$(1+\beta) \eta n \leq \delta \leq \frac{1}{2}$
 $\Rightarrow n \leq \frac{1}{(1+\beta) \eta n} \leq 1$

because we need all i to satisfy this bound so, prob $\geq \frac{\delta}{n}$

$$\text{So, } R_T \leq \beta n T + \sigma T + (1+\beta)\gamma n T + \frac{\ln(\delta^{-1}n)}{\beta} + \frac{\ln n}{\gamma}$$

now, plug in the values, separating between the trivial $T \leq 5.15 \sqrt{T n \ln(\delta^{-1}n)}$ and large T ($\Rightarrow \sigma \leq 0.21$ and $\beta \leq 0.1$, hence the required ineq.)

H.W. (1) complete the argument
(2) prove that with same values

$$E[R_T] \leq 5.15 \sqrt{T n \ln n} + \sqrt{\frac{T n}{\ln n}}$$

One can show a lower bound of $\Omega(\sqrt{T n})$

1. Consider the experts setting with gains: $g_{i,t} \in [0, 1]$ is the gain of expert i at step t .

Hedge updates

$$p_{i,t+1} = \frac{e^{\eta G_{i,t}}}{\sum_j e^{\eta G_{j,t}}}$$

where $G_{i,t} = \sum_{s \leq t} g_{i,s}$

Prove that the regret of Hedge at time T is $O(\sqrt{T \log n})$, for a good choice of the learning rate η , against the adaptive adversary.

2. Show a lower bound of $\Omega(\sqrt{T})$ in the experts setting on the regret of any online algorithm against the oblivious adversary.

3. Consider a system of linear inequalities $Ax \geq b$, where $A \in [0, \infty)^{m \times n}$, $b \in [0, \infty)^m$, and the unknown $x \in [0, \infty)^n$ (we're seeking a non-negative solution). An ε -approximate solution $x \geq 0$ satisfies $Ax \geq b - \begin{pmatrix} \varepsilon \\ \vdots \\ \varepsilon \end{pmatrix}$. Suppose we have an efficient procedure for following problem: Given $p \in [0, 1]^m$, $\sum_{i=1}^m p_i = 1$, decide if $\exists x \geq 0$, $p^T A x \geq p^T b$. Show how to find an ε -approx. solution to $Ax \geq b$. Analyze the run-time.

4. Recall that we showed, for EXP3.P, that, w.p. $\geq 1 - \delta$:

$\swarrow$ time T regret

$$R_T \leq \beta n T + \delta T + (1 + \beta) \eta n T + \frac{\ln(\delta^{-1} n)}{\beta} + \frac{\ln n}{\eta}$$

Infer that for the right choice of β, δ, η ,

$$E[R_T] = O(\sqrt{T n \ln n})$$